Physics 101: Mechanics

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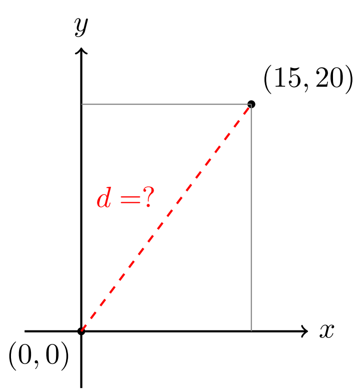
Contents

Chapter 1

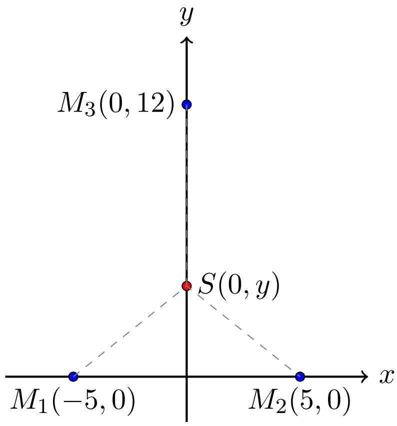
The Distance Formula using pythagoras

EXPERIMENTAL LAB: FIELD APPLICATIONS

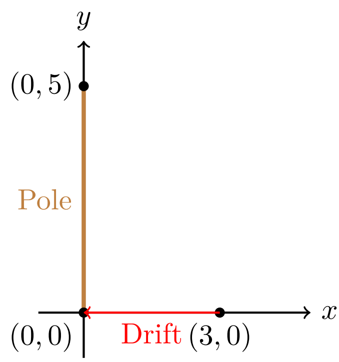
1. **LaserRangefinderCalibration:** A surveyor places a laser at origin $(0, 0)$ and a reflector at $(15, 20)$. The laser reads 24.8 units. Calculate the theoretical distance and find the percentage error.



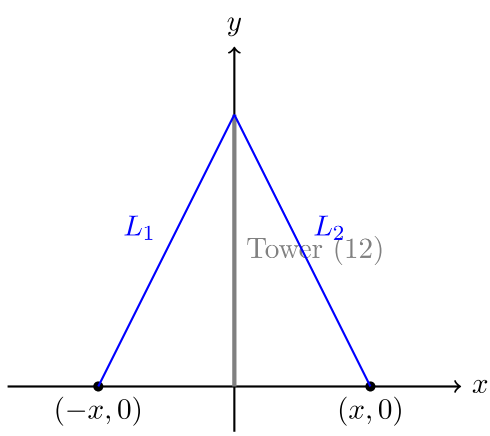
2. **The Acoustic Localization:** Two microphones at $M_1(-5, 0)$ and $M_2(5, 0)$ detect sound. A third at $M_3(0, 12)$ detects it later. Find the source $S(0, y)$.



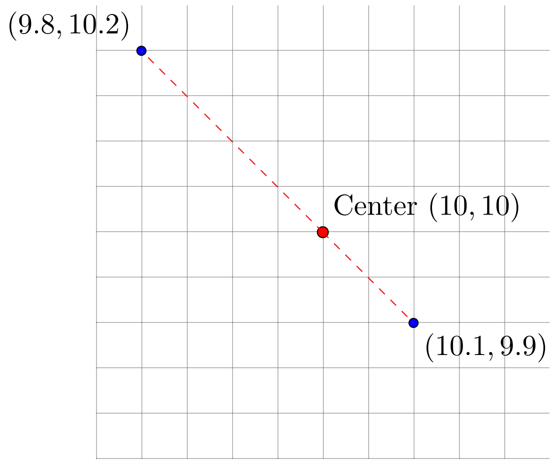
3. **Shadow Tracking:** A pole's top is at $(0, 5)$ and its shadow is at $(3, 0)$. When the sun moves, the shadow moves to $(0, 0)$. Calculate the total distance the shadow tip traveled.



4. **Tension Wire Stability:** A tower at $(0, 12)$ is secured by wires anchored at $(x, 0)$ and $(-x, 0)$. If the total length of both wires is 26, find x.



5. **GPS Drift Analysis:** Readings at $(10, 10)$ drift to $(10.1, 9.9)$ and $(9.8, 10.2)$. Find the average distance of these drift points from the center.


(9.8, 10.2)
Center (10, 10)
(10.1, 9.9)

Chapter 2

Coordinate Geometry

2.1 Introduction

Coordinate geometry, also known as analytic geometry, is the study of geometry using a coordinate system. This contrasts with synthetic geometry.

2.2 The Cartesian Plane

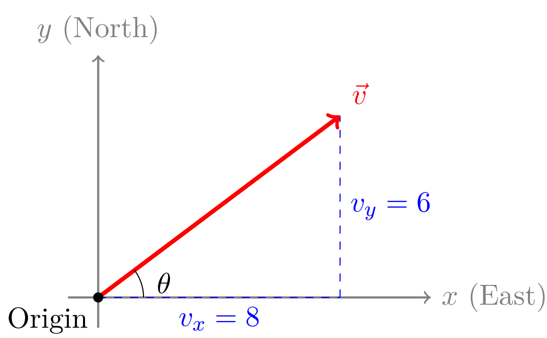
The Cartesian plane is defined by two perpendicular number lines: the x-axis, which is horizontal, and the y-axis, which is vertical.

Chapter 3

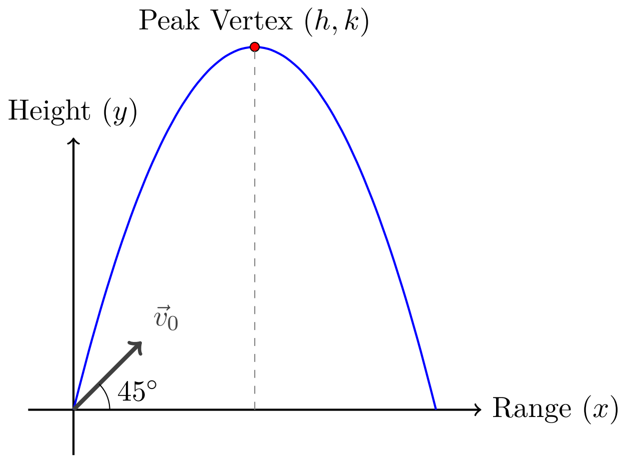
New Chapter

EXPERIMENTAL LAB: VECTOR KINEMATICS & DYNAMICS

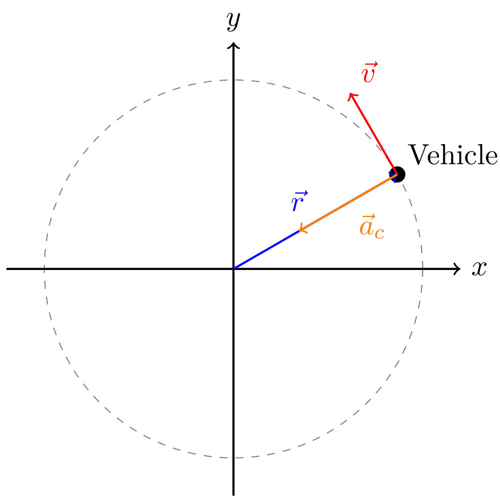
1. **Drone Navigation Vectors:** A quadcopter flies from a launch pad with a velocity vector $\vec{v} = 8\hat{\imath} + 6\hat{\jmath}$ meters per second. Calculate its total ground speed $|\vec{v}|$ and the heading angle θ relative to the positive x-axis.



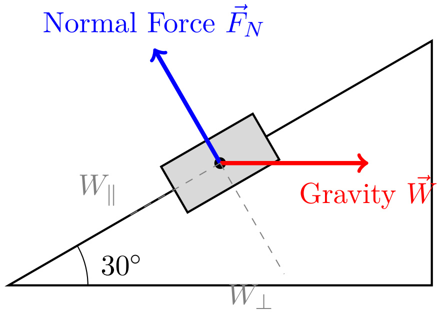
2. **Projectile Trajectory Analysis:** A ball launcher fires a projectile from ground level at an angle of 45° with an initial speed of 14 m/s. Sketch the parabolic path and compute the peak vertex (h, k) where vertical velocity $v_y = 0$.



3. **Centripetal Force Boundaries:** A tethered test vehicle undergoes uniform circular motion around a central pillar at $(0, 0)$ with a fixed radius of 5 meters. Show the direction of the centripetal acceleration vector $\vec{a}_c$ relative to the position vector $\vec{r}$.



4. **Static Inclined Plane Equilibrium:** A block weighing 20 N rests on a friction-less ramp inclined at 30°. Draw the complete Free-Body Diagram (FBD) breaking down the gravity vector $\vec{W}$ into parallel and perpendicular components.


Normal Force $\vec{F}_N$
$W_{\parallel}$
Gravity $\vec{W}$
30°
$W_{\perp}$